

Determinant and Zero-Mode Quotients on a Frontier Lift: Kernel Equivalence and the Missing Canonical-Parent Bridge

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

The determinant fibers used by the TPC energy gate and the ordered outer fibers used by the signed zero-mode gate are two different quotients of a literal occurrence archive. We give an exact simultaneous-lift criterion. For finite surjective maps Q_D, Q_Z with a common domain, an isomorphism J satisfying $JQ_D = Q_Z$ exists precisely when $\ker Q_D = \ker Q_Z$. If J must be a literal relabeling, equality of kernels is supplemented by equality of occurrence partitions and exact map multipliers. Equality on one coefficient vector, or of one final scalar sum, does not imply this criterion. Applying the test to the source-locked cut archive exposes the exact missing fields. The determinant quotient needs the canonical parent, targets, content, normalized determinant and inverse aggregation; the zero quotient needs the outer affine key, canonical order, arithmetic sign, reconstruction weight and content allocation. Neither field family is connected to the cut paths by the absent occurrence lift. Thus the abstract quotient test is proved at L0, whereas both literal totality statements and their intertwining remain NOT-TESTABLE. No determinant-energy, zero-mode or fixed-shift arithmetic exponent is claimed.

1 Two quotients, one literal domain

Let $\mathcal{O}_X$ be a fully typed row-separated occurrence space on all nonsoft cut paths. This includes both eligible-tail-open and frontier-unmapped records. TPC-143 identifies the still missing conservative lift from the cut archive to $\mathcal{O}_X$ [5, 6].

The determinant construction of TPC-95 and TPC-110 first inverse-aggregates computational children to canonical mathematical parents $t = (\alpha, \gamma, j)$. A parent has ordered slopes m_α, m_γ , targets

$$x_t = m_\alpha j + h_0, \quad y_t = m_\gamma j + h_0,$$

content $c_t = (x_t, y_t)$, and the literal determinant label

$$\nu_D(t) = \frac{m_\alpha - m_\gamma}{c_t} \in \mathbb{Z} \setminus \{0\}. \quad (1)$$

We write ν_D rather than ν to prevent collision with the physical normalization identifier ν_{phys} .

The zero-mode construction of TPC-111 and TPC-122 instead uses outer affine keys θ , canonically ordered coordinates

$$r_{\theta,1} < \cdots < r_{\theta,m_\theta},$$

literal signs $\sigma_{\theta,i}$, outer weights $w_{\theta,i}$, and an exact retained/content allocation [1, 3]. The order is part of the object: permuting a fiber can preserve its final scalar while changing every signed-prefix maximum.

Definition 1.1 (Literal quotient maps). Let $H_X = \mathbb{C}^{\mathcal{O}_X}$. A literal determinant quotient

$$Q_D : H_X \longrightarrow K_D$$

is the exact inverse-aggregation and determinant-binning map on the canonical parents. A literal zero-mode quotient

$$Q_Z : H_X \longrightarrow K_Z$$

is the exact outer-record aggregation with its canonical order and factor allocation retained in the output records.

The linear maps contain exact multipliers; a target label without its edge multiplier is not a quotient implementation. A zero literal coefficient is not permission to omit its formal occurrence unless an exact aggregation theorem supplies that zero.

2 The quotient-kernel criterion

Theorem 2.1 (Simultaneous quotient lift). *Let H, K_D, K_Z be finite-dimensional vector spaces and let*

$$Q_D : H \rightarrow K_D, \quad Q_Z : H \rightarrow K_Z$$

be surjective. There exists a unique isomorphism $J : K_D \rightarrow K_Z$ satisfying

$$JQ_D = Q_Z \tag{2}$$

if and only if

$$\boxed{\ker Q_D = \ker Q_Z.} \tag{3}$$

Proof. If (2) holds with J injective, then

$$Q_D v = 0 \iff JQ_D v = 0 \iff Q_Z v = 0.$$

Conversely, under (3) define

$$J(Q_D v) = Q_Z v.$$

If $Q_D v = Q_D w$, then $v - w \in \ker Q_D = \ker Q_Z$, so the definition is well posed. It is linear and onto because Q_Z is onto. Its kernel is zero by the same kernel equality, hence it is an isomorphism. Surjectivity of Q_D makes J unique. $\square$

Corollary 2.2 (Row-space execution). *After bases are fixed, (3) is equivalent to equality of the row spaces of Q_D and Q_Z . It is therefore decidable by exact row reduction.*

Proof. For a finite matrix A , the row space is the annihilator of $\ker A$. Equal kernels have equal annihilators, and conversely. $\square$

Remark 2.3 (Literal relabeling is stronger). The isomorphism in [theorem 2.1](#) is an abstract linear map. If the proposed J is a relabeling of literal fibers, the occurrence equivalence relations, canonical orders where applicable, and exact map multipliers must also match. Equal kernels do not authorize an unrecorded change of physical meaning.

Proposition 2.4 (Literal relabeling test). *Fix occurrence-column orderings and typed bases of K_D, K_Z . For an allowed metadata-preserving bijection φ between the two output-record sets, let Π_φ be its permutation matrix. Then φ intertwines the literal quotients if and only if*

$$Q_Z = \Pi_\varphi Q_D.$$

Equivalently, after the allowed relabeling, the complete weighted rows —not merely their supports or final scalar sums—agree exactly. Kernel equality alone does not imply this condition.

Proof. The displayed equality is precisely the coefficientwise statement $\Pi_\varphi Q_D v = Q_Z v$ for every occurrence vector v . For strictness, take

$$Q_D = I_2, \quad Q_Z = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Both kernels are zero, but the rows of Q_Z are not a permutation of the rows of Q_D . $\square$

3 Why scalar equality is insufficient

Proposition 3.1 (A conservative scalar can hide distinct kernels). *There are surjective incidence quotients Q_D, Q_Z such that*

$$\mathbf{1}^T Q_D c = \mathbf{1}^T Q_Z c \quad \text{for every } c,$$

but $\ker Q_D \neq \ker Q_Z$.

Proof. Take

$$Q_D = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q_Z = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Both matrices have column sums one, so their final scalar sums agree on every vector. But

$$(1, -1, 0) \in \ker Q_D, \quad Q_Z(1, -1, 0)^T = (1, -1)^T \neq 0.$$

□

TPC-124 already requires the coefficientwise matrix identity $(JQ_D - Q_Z)M = 0$, not agreement on one literal coefficient vector [4]. The kernel criterion explains the quotient geometry behind that firewall. Even equality for every final scalar does not determine which occurrences were identified. If M is not surjective onto H_X , however, the TPC-124 identity constrains only $\text{Ran } M$. Then the full-space kernel equality in [theorem 2.1](#) is sufficient but need not be necessary; the exact restricted criterion compares

$$\ker(Q_D|_{\text{Ran } M}) \quad \text{and} \quad \ker(Q_Z|_{\text{Ran } M})$$

with codomains replaced by their respective images. A full-space necessity claim therefore requires a proved surjectivity statement for M . This restricted kernel test concerns existence of an abstract image isomorphism; a prescribed or literal J must still pass the coefficientwise identity on $\text{Ran } M$.

4 Exact field requirements

Before either quotient is compared, both matrices must provide the same ordered occurrence-column registry. Each must also certify surjectivity onto its declared codomain or explicitly replace that codomain by the image used in the restricted criterion.

For Q_D , every completed occurrence lineage must provide:

- D1.** the canonical parent and exact child-to-parent aggregation;
- D2.** row identifiers, integer slopes, j , targets and content;
- D3.** an exact verification of (1);
- D4.** physical and computational multiplicities in distinct fields;
- D5.** the literal parent coefficient u_t ; and
- D6.** the determinant-bin target and exact map multiplier.

For Q_Z , it must provide:

- Z1.** the outer affine key and ordered coordinate;
- Z2.** the canonical rank within the outer interval;
- Z3.** the literal arithmetic sign and outer reconstruction weight;
- Z4.** the fixed factor-allocation identifier;
- Z5.** retained/content status and content-remainder destination; and

Z6. the zero-mode output record and exact map multiplier.

These lists are data contracts, not determinant or prefix estimates. In particular, recording u_t does not prove phase coherence, and recording $\sigma_{\theta,i}$ does not prove a signed-prefix saving [2, 3].

Proposition 4.1 (The common parent bridge is absent). *The committed TPC-136 and TPC-143 artifacts contain no exact map from a nonsoft cut path to the canonical parent or outer occurrence records above. Consequently neither literal quotient matrix can be formed from the current artifacts.*

Proof. TPC-143 performs an exact field census on every nonsoft cut record. The cut records contain the native tuple (ℓ, k, d) , dyadic block, cut type, h_0 and normalization. They contain none of the canonical-parent, inverse-aggregation, outer-key, ordered-coordinate or quotient-edge fields listed above. A statement defining those objects in an earlier paper is not a path-ID crosswalk. Thus the matrix rows and columns required for Q_D, Q_Z have no current literal source. $\square$

5 Executable theorem and actual verdict

The companion program `experiments/tpc144_quotient_kernel_audit.py` uses exact rational row reduction. Synthetic matrices verify [theorem 2.1](#) and [propositions 3.1](#) and [2.4](#), including the surjectivity precondition; they are explicitly marked `SYNTHETIC_L0_ONLY`. A separate actual manifest imports the TPC-143 occurrence-lift status and contains no fabricated map edge. Default mode writes deterministic JSON and `-check` compares it byte for byte without writing.

Statement	Status	Scope
Surjective quotient kernel criterion	PROVED _{L0}	Exact finite-dimensional theorem.
Literal Q_D totality	NOT-TESTABLE	Missing occurrence and canonical-parent lift.
Literal Q_Z totality	NOT-TESTABLE	Missing outer records, order and allocation.
Q_D/Q_Z intertwining	NOT-TESTABLE	Neither actual kernel is available.
Schema/scalar-only derivation	STOP-DECLARED-ROUTE	Scoped stop; augmented lift remains open.

The audit rejects a frontier-only domain, a fabricated quotient edge, a proved status without a source, deletion of a required D6/Z6 field, promotion of either actual kernel or literal relabeling, and a scalar-to-coefficientwise promotion. Its required domain remains

$$\mathcal{C}_X^{\text{ETO}} \sqcup \mathcal{C}_X^{\text{FUM}},$$

even though the current finite sample has no eligible-tail row.

6 Claim boundary

Theorem 6.1 (Current quotient verdict). *For the current source-locked occurrence obligations,*

$$\begin{aligned} \text{H1.frontier_QD_totality} &= \text{NOT-TESTABLE}, \\ \text{H1.frontier_QZ_totality} &= \text{NOT-TESTABLE}, \\ \text{H1.frontier_QD_QZ_intertwining} &= \text{NOT-TESTABLE}. \end{aligned}$$

The direct schema-or-scalar derivation is stopped, while a theorem-backed occurrence-augmented route is not stopped.

No assertion here bounds the complete frontier scalar, proves that a formal coefficient is nonzero, lower-bounds determinant energy, controls the distinguished zero mode, gives a positive fixed- h_0 L2 saving, pays the 1/400 endpoint, or proves a prime-pair or twin-prime statement. The next input remains the literal occurrence lift identified by TPC-143.

References

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